

CASE REPORT

Toxicology

Forensic pathology and toxicological analyses: A case of fatal xylazine poisoning

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Abstract

Xylazine, an α_2 -adrenergic receptor agonist commonly used as a veterinary sedative, has gained notoriety for its increasing misuse in combination with opioids, such as fentanyl, to potentiate its effects. This trend has led to a rise in cases of xylazine poisoning. Although timely diagnosis and intervention have reduced fatalities, the post-mortem pathological manifestations and lethal blood concentration of xylazine alone remain poorly characterized due to its common co-occurrence in toxicology with fentanyl. This study presented a case of homicide involving fatal intoxication solely from xylazine overdose without coadministration of other central nervous system drugs, detailing the multi-organ pathological changes observed and quantifying the decedent's blood xylazine concentration at 13.34 $\mu\text{g/mL}$ using liquid chromatography-tandem mass spectrometry (LC-MS/MS). The findings aimed to provide forensic practitioners with valuable insights for the identification and investigation of fatal xylazine poisoning cases.

KEYWORDS

alpha-2-adrenergic agonist, forensic pathology, forensic toxicology, homicide, liquid chromatography-tandem mass spectrometry and LC-MS/MS, xylazine poisoning

Highlights

- A homicidal case of xylazine-only poisoning was reported.
- The pathological changes of vital organs were presented in detail.
- A LC-MS/MS method for quantifying xylazine concentration in blood was optimized and successfully applied to this case.

1 | INTRODUCTION

Xylazine, a potent α_2 -adrenergic receptor agonist, is extensively utilized in veterinary medicine for its sedative, analgesic, and muscle relaxant properties [1]. However, its pharmacological effects have also led to increasing misuse in combination with opioids, such as fentanyl, to potentiate its properties [2]. This trend has resulted in a growing number of xylazine-related poisoning cases, particularly

within the context of substance abuse. Despite its rising prevalence, xylazine poisoning remains inadequately characterized in forensic medicine, with limited data on its pathological mechanisms, post-mortem findings, and lethal concentrations in humans [3].

In recent years, sporadic cases of xylazine poisoning have been documented, frequently linked to accidental overdoses or intentional misuse [4]. Although prompt medical intervention has lowered mortality rates, fatalities persist, particularly when xylazine is

coadministered with other central nervous system (CNS) depressants. Documented cases of fatal intoxication from xylazine alone remain exceptionally rare in the existing literature [5]. The scarcity of comprehensive data on postmortem pathological alterations and toxicological thresholds for xylazine presents significant challenges for forensic pathologists in accurately determining the cause and manner of death in such cases [6].

This article reports a homicide case involving xylazine overdose, detailing the distinctive multi-organ pathological alterations identified during autopsy. Furthermore, the decedent's blood xylazine concentration was quantified using the LC-MS/MS method, providing a definitive lethal reference value for isolated xylazine intoxication. Through this case report, we aim to augment the limited forensic literature on xylazine poisoning, providing essential reference data for the diagnosis and investigation of similar cases in the future.

2 | CASE REPORT

2.1 | Case history

A couple agreed to commit suicide at home using a xylazine solution purchased online. The woman initially injected one dose of the solution into the dorsal hand veins of both herself and the male individual. Shortly after administration, both individuals lost consciousness and fell into a coma. Approximately 6 h later, the woman regained consciousness and observed that the man exhibited weak respiratory activity. She then administered an additional five doses of the solution into the muscles of his neck. About 15 h later, noting that the man still displayed faint respiratory efforts and possibly experiencing regret over their suicide pact, she contacted emergency services. Despite resuscitation attempts, the man ultimately succumbed to the overdose. Based on the manufacturer's specified concentration (0.2 g/2 mL) and the reported six doses, the estimated dose administered to the decedent was calculated to be approximately 1.2 g of xylazine during the final 21 h before death (The packaging boxes of xylazine, the used syringes, and the empty vials of xylazine (fully consumed) were shown in Figure S1). An autopsy was performed 11 days postmortem, during which the body had been preserved in a frozen state.

2.2 | Autopsy and histopathological findings

The frozen male cadaver (168.0 cm in length) exhibited the following pathological features: *Left-hand dorsum*: A 2.0 cm × 1.5 cm area of cutaneous bruising containing a single needle puncture site. Histological examination revealed multifocal subcutaneous hemorrhages with sparse inflammatory infiltrates. *Left sternocleidomastoid muscle*: Eleven needle puncture sites clustered within a 3.0 cm × 2.5 cm area. Incision demonstrated subcutaneous soft tissue hemorrhage (3.0 cm × 2.0 cm). Microscopically, patchy subcutaneous hemorrhages were observed, accompanied by dense erythrocyte

extravasation and a neutrophilic-predominant mixed inflammatory infiltrate. *Right sternocleidomastoid muscle*: Six needle puncture sites within a 3.0 cm × 1.5 cm area. Subcutaneous hemorrhage (2.0 cm × 2.0 cm) and deep muscular hemorrhage (4.0 cm × 3.0 cm) were identified upon dissection. Histological analysis showed multifocal subcutaneous hemorrhages with localized hematoma formation and a neutrophilic-rich inflammatory infiltrate. *Brain*: The brain (weight: 1415 g) demonstrated histopathological features consistent with ischemic-hypoxic encephalopathy, characterized by neuronal hyperchromasia and neuronophagia evidenced by microglial infiltration into neuronal cytoplasm. Glial cell proliferation was notably increased (Figure 1A). *Lungs*: The lungs (combined weight: 1380 g) exhibited multifocal hemorrhagic lesions, measuring 7.0 cm × 7.0 cm (left lower lobe) and 1.0 cm × 1.0 cm (right middle lobe). Histological examination revealed patchy alveolar hemorrhages with mild inflammatory infiltrates. Alveolar spaces contained eosinophilic edema fluid, accompanied by emphysematous changes and marked vascular congestion (Figure 1B). *Liver*: The liver (1450 g) showed focal coagulative necrosis, evidenced by hepatocyte nuclear fragmentation, cytoplasmic eosinophilia, and disruption of hepatic cord architecture. Binucleated hepatocytes were present, with significant vascular congestion and ice crystal artifacts in interstitial spaces (Figure 1C). *Kidneys*: The kidneys (combined weight: 260 g) demonstrated mild glomerular hyalinization. Proximal tubular epithelial cells exhibited features of coagulative necrosis, including nuclear dissolution and cytoplasmic eosinophilia (Figure 1D). No significant pathological alterations were observed in other organs.

2.3 | Toxicological analyses

Postmortem blood samples from IVC were analyzed for xylazine concentration using a validated LC-MS/MS method.

2.3.1 | Sample preprocessing

100 µL aliquots of whole blood was spiked with 10 µL of xylazine-d₆ internal standard solution (100 ng/mL) and protein precipitated with 900 µL of MS-grade acetonitrile (SCRC, China). Following vortex mixing (1 min) and centrifugation (700 g, 1 min; DLAB, China), the supernatant was filtered through a 0.22-µm membrane (Beyotime, China). A 5 µL aliquot of the processed sample was injected into the LC-MS/MS system for quantitative analysis.

2.3.2 | LC conditions

Chromatographic separation was performed using an LC-MS/MS 8045 system (SHIMADZU, Japan) equipped with a C18 column (150 mm × 3 mm, 5 µm, Phenomenex Gemini, USA). The mobile phase consisted of: (A) 0.1% ammonium acetate (MS grade, ALADDIN, China) and (B) methanol (MS grade, SCHARLAU, Spain). The column

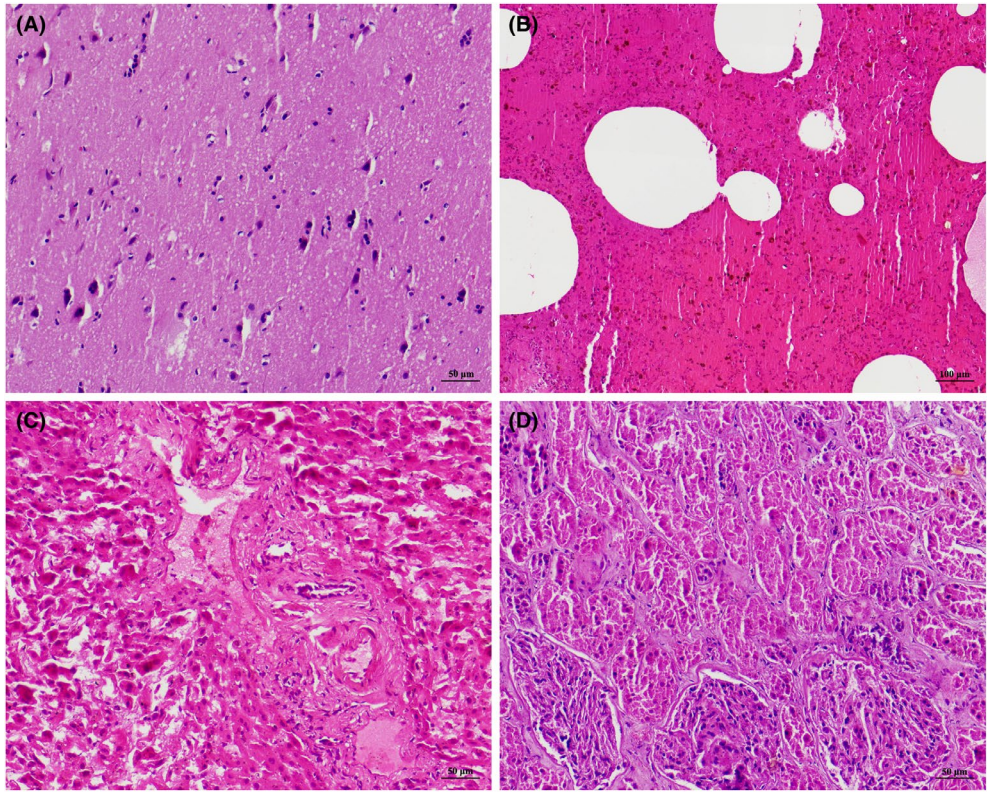


FIGURE 1 Positive pathological changes in vital organs (A: Ischemic-hypoxic changes in neurons; B: Pulmonary hemorrhage, congestion, edema, and emphysema; C: Coagulative necrosis of hepatocytes; D: Necrosis of renal tubular epithelial cells).

	Precursor ion (m/z)	Product ion (m/z)	Q1 pre bias (V)	CE	Q3 pre bias (V)
Xylazine	221.10	90.10	-13	-23	-15
	221.10	164.10	-13	-26	-30
Xylazine-d6	227.15	90.10	-11	-24	-16
	227.15	170.10	-10	-27	-16

TABLE 1 Some significant MRM parameters for the analysis of xylazine.

Note: The ions in bold were used for quantification purposes.

temperature was maintained at 30°C, and the flow rate was set at 0.35 mL/min. The gradient elution program was as follows: initial conditions of 10% B, increased linearly to 95% B over 3 min, held for 3 min, then returned to 10% B in 0.1 min, and re-equilibrated for 5.9 min prior to the next injection. The total run time was 12 min.

2.3.3 | MS/MS conditions

Mass spectrometric detection was performed using an electrospray ionization (ESI) source in positive ionization mode with multiple reaction monitoring (MRM). The instrument parameters were optimized as follows: dwell time, 50 ms; heating gas (air) flow rate, 10 L/min; nebulizing gas (nitrogen) flow rate, 3 L/min; drying gas (nitrogen) flow rate, 10 L/min; interface temperature, 300°C; interface voltage, 4 kV; desolvation line (DL) temperature, 250°C; and heat

block temperature, 400°C. Additional MRM parameters, including precursor/product ion transitions and collision energies, are detailed in Table 1.

2.3.4 | Method validation

Method validation for LC-MS/MS analysis was systematically conducted in blank blood samples using isotope-labeled standards. The evaluated parameters included specificity, limit of detection (LOD), limit of quantification (LOQ), linearity, matrix effect, extraction recovery, accuracy, precision, dilution reliability, and carryover.

1. Specificity: Blank blood samples and blank blood samples spiked with internal standards from 10 different donors were subjected to pretreatment. No interfering substances were detected at

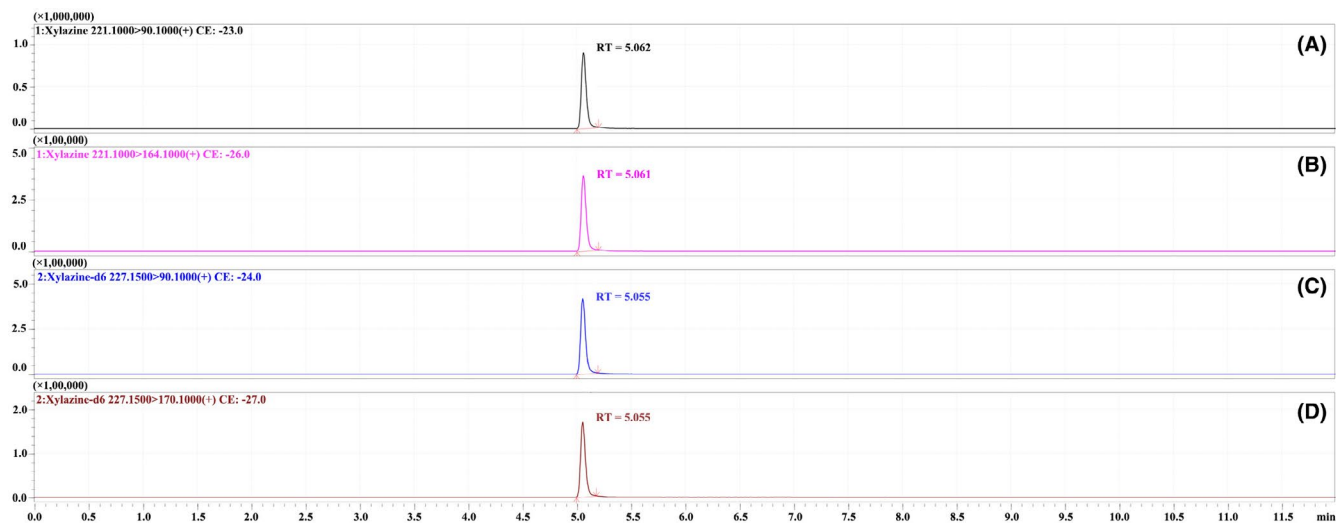


FIGURE 2 The chromatogram of xylazine in the diluted blood sample (A: Xylazine, 221.10 > 90.10; B: Xylazine, 221.10 > 164.10; C: Xylazine-d6 as internal standard, 227.15 > 90.10; D: Xylazine-d6 as internal standard, 227.15 > 170.10).

the retention time of xylazine, demonstrating excellent method specificity.

2. LOD and LOQ: Three concentration levels (1, 5, and 10 ng/mL) were tested to determine LOD and LOQ. No signal was detected at 1 ng/mL. The 5 ng/mL concentration demonstrated a signal-to-noise (S/N) ratio > 3 (LOD), while 10 ng/mL showed S/N > 10 (LOQ).
3. Linearity: A seven-point calibration curve (10, 20, 50, 100, 500, 1000, and 1500 ng/mL) demonstrated excellent linearity with a correlation coefficient (r) of 0.9996, while the regression equation was determined as $Y = 0.0122245X - 0.115527$.
4. Matrix effect and extraction recovery: The matrix effects at 50 and 1000 ng/mL were determined to be 97.1% (RSD = 5.5%) and 98.7% (RSD = 4.8%), respectively, with corresponding extraction recoveries of 96.0% (RSD = 4.6%) and 101.0% (RSD = 5.3%).
5. Accuracy and precision: The accuracy and precision of four concentrations (10, 20, 100, and 1000 ng/mL) were measured. Results showed that intraday accuracy ($n = 6$) ranged from 89.01% to 97.59% with precision (RSD) of 3.6%–8.62%; interday accuracy ($n = 15$) was in the range of 90.12%–96.83% with precision (RSD) of 4.1%–7.9%.
6. Dilution reliability: A sample spiked at 10 times the upper limit of quantification (15,000 ng/mL) was diluted 10-fold with blank blood matrix prior to analysis. The diluted sample demonstrated an accuracy of 96.6% and precision (RSD) of 4.2%, confirming reliable quantitation at extended concentration ranges.
7. Carryover: Carryover was assessed by analyzing a blank sample immediately following the concentrations of 10, 100, and 1500 ng/mL. The requirement was that the analyte peak area in the subsequent blank was less than 20% of the peak area of the limit of quantification sample. The results demonstrated that no carryover effect was observed at 10 and 100 ng/mL, while values of 1.52%–3.17% (relative to LOQ) at 1500 ng/mL confirmed an acceptable method carryover effect.

2.3.5 | LC-MS/MS results

Following qualitative identification of xylazine in blood samples, preliminary assessment indicated analyte concentrations exceeding the method's calibration range (10–1500 ng/mL). To enable accurate measurement, a validated 10-fold dilution protocol using analyte-free blood matrix was implemented. Post-dilution quantitative analysis yielded a concentration of 1.33 μ g/mL (Figure 2), corresponding to an original undiluted concentration of 13.34 μ g/mL after applying the dilution factor (10 $\times$). No ethanol or other common toxic substances were identified in the analyzed samples (the results are shown in Figure S2).

2.4 | Forensic diagnosis

Based on the integration of case history, autopsy findings, histopathological examination, and toxicological analysis, the cause of death was determined to be acute xylazine poisoning.

3 | DISCUSSION

This study presented a rare case of fatal xylazine poisoning, documenting comprehensive histopathological alterations in multiple organs, a validated toxicological analysis method, and quantitative results. These findings provided critical insights into the toxic effects and potential lethal mechanisms of xylazine, contributing significantly to the forensic identification of xylazine-related fatalities [7, 8]. In this case, the decedent exhibited multisystem pathological changes, including: (1) subcutaneous hemorrhage at the injection site; (2) neuronal ischemic-hypoxic injury; (3) pulmonary hemorrhage, edema, and emphysema; (4) hepatic coagulative necrosis; and (5) renal tubular epithelial cell necrosis. These

findings were consistent with the systemic toxic effects of xylazine, highlighting its potential to induce multi-organ dysfunction and failure.

First, despite the relatively short intoxication period (approximately 21 h), the brain exhibited characteristic neuronal necrotic changes, including nuclear pyknosis, cytoplasmic eosinophilia, and glial cell infiltration with neuronophagia. These histopathological features, rarely documented in the literature [9, 10], suggested rapid and severe neurotoxic effects of xylazine. Second, pulmonary pathological manifestations including hemorrhage, edema, and emphysema indicated significant respiratory suppression during the intoxication period. The combined neuropathological and pulmonary findings suggested that xylazine overdose induces profound suppression of the central nervous and respiratory systems, leading to cerebral and pulmonary hypoxia [11, 12]. Furthermore, the presence of hepatic coagulative necrosis and renal tubular epithelial cell necrosis implied that xylazine may exert direct cytotoxic effects on the liver and kidneys, consistent with toxic hepatopathy and nephropathy. These observations align with previous studies [13, 14], supporting the systemic toxicity of xylazine.

In addition to the pathological findings, quantitative analysis of xylazine in postmortem blood was conducted using liquid chromatography-tandem mass spectrometry (LC-MS/MS) [15–17]. The results revealed a markedly elevated xylazine concentration of 13.34 µg/mL, which significantly exceeded the toxic and lethal blood concentrations reported in prior studies, such as 0.057 µg/mL [18], 0.105 µg/mL [19], and a range of 0.005–0.2 µg/mL in 55 cases [6]. This finding provided critical evidence supporting the conclusion that the cause of death was acute xylazine intoxication.

However, this study had several limitations, including its reliance on a single case, the absence of large-sample data, and the lack of detection of xylazine metabolites. So, we thought that future research should focus on animal experiments or the collection of additional case studies to further elucidate the toxic mechanisms of xylazine and its metabolites, as well as interindividual metabolic variations. Such investigations will enhance the understanding of xylazine toxicity and provide more robust guidance for clinical and forensic practices.

4 | CONCLUSIONS

In conclusion, given the increasing global misuse of xylazine and the rarity of fatal xylazine intoxication cases, we hope that the multi-organ pathological changes, postmortem blood detection method for xylazine, and the elevated blood concentration (13.34 µg/mL) reported in this case may provide valuable reference data for forensic identification in similar cases.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Yu Z, Yuan Y, Zhang J, Gao X, Chen C, Zhao E, et al. Forensic pathology and toxicological analyses: A case of fatal xylazine poisoning. J Forensic Sci. 2025;70:2102–7. <https://doi.org/10.1111/1556-4029.70104>